

Prediction of Heavy Metal Concentrations (Pb, Cd) in Precipitation Using Machine Learning

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1 Introduction

This experiment investigates the prediction of heavy metal concentrations (Lead (Pb) and Cadmium (Cd)) in precipitation samples using machine learning (ML) models trained on environmental data. The data set consists of precipitation samples collected in various location across Austria, characterised by the total amount of precipitations, several ion concentrations, and conductivity. In particular, the coverage for Pb and Cd for certain observation stations is incomplete or missing, sine they require dedicated instrumentation. Given the importance of these two elements for they dangerous polluting properties, it is valuable implement algorithmic ML methods to predict their quantity based on the other characteristics of the samples.

2 Data Description

The data set Redl, Steinkogler and Kasper-Giebl 2021 in use can be retrieved on the [TU Wien Research Data Portal 2026](#) portal. The data set is shared under the license [Creative Commons Attribution Non Commercial Share Alike 4.0 International 2026](#), granting the usage for the scope of this study. The data set comprises two distinct CSV files named *precipitationdata.csv* and *stationcoordinates.csv*, containing the actual chemical and physical characterization and the information (name, latitude and longitude) of the collection locations, respectively. The data set is structured upon the following 15 columns:

- *Datum, Ort* - date and location of the precipitation
- *NS* - precipitation amount in *nm*
- *LF* - conductivity in $\mu S/cm$
- *pH* - pH value
- NH_4 , Na, K, Ca, Mg, Cl, NO_3 , SO_4 - ions concentrations in *mg/l*
- Pb, Cd - concentration in $\mu g/L$

The data set has a total time range of around 5 years, with the first precipitation dating October 1st 2014 and the last dating December 30th 2020, for a total of 8572 data points spread across the various stations.

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3 Implementation

The experiment is implemented as a reproducible ML pipeline comprising data preprocessing, feature selection, dataset splitting into training/test/validation, model training, and evaluation of the most common metrics. The work flow is implemented using standard *python* coding language and libraires, in the form of *jupyter* notebooks, to facilitate code comprehension and clear reproducibility of the results.

The input data set is first split between the observations lacking the Pb and Cd measurements, which represent our prediction target, and the remaining to be used for our model training which is then split into training (70%), validation (15%), and test (15%).

Two independent regression models are trained for Pb and Cd using Linear Regression and Random Forest. We aim at testing two different approaches: for the first one, we predict separately the values of the missing Pb and the missing Cd using only the remaining parameters. The second approach uses the multi variables prediction feature of the *RandomForestRegressor* in *sklearn* which predicts simultaneously Pb and Cd.

Various intermediate results, including statistical visualizations, correlation matrices, model performance comparisons, and predictions are created. The model is then used for the predictions of the missing Pb and Cd concentrations. Diagram Flow of the experiment can be visualized in 1.

All the workflow implemented in the code supports good data stewardship practices i.e. data source, reproducibility, traceability, and proper data management.

4 Diagram Flow

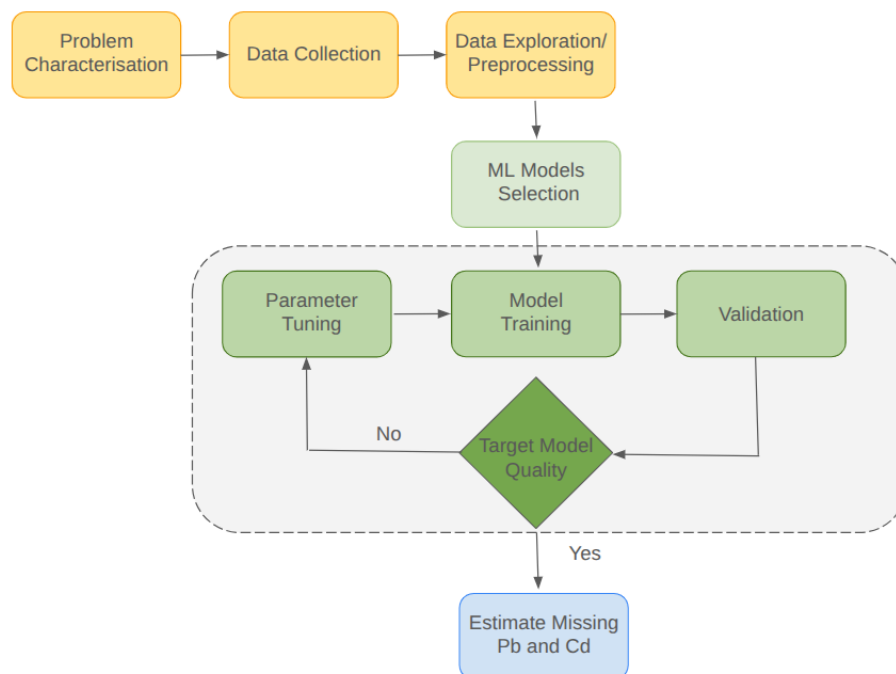


Figure 1: Diagram Flow of the ML Pb and Cd prediction Experiment.

References

- Creative Commons Attribution Non Commercial Share Alike 4.0 International* (2026). <https://creativecommons.org/licenses/by-nc-sa/4.0/deed.en>.
- Redl, Peter, Thomas Steinkogler and Anne Kasper-Giebl (july 2021). "Concentrations of major ions in wet precipitation samples in Austria". **version** 1.0.0. **in**Accessed: 2026-03-23: DOI: [10.48436/b0g4h-rv840](https://doi.org/10.48436/b0g4h-rv840).
- TU Wien Research Data Portal* (2026). <https://researchdata.tuwien.ac.at>.